RETAIL STORE SALES INSIGHTS AND PREDICTION MODEL

PHASE 1

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1. INTRODUCTION

This project aims to perform an in-depth analysis of retail store sales data to identify key trends, patterns, and performance indicators that can assist in strategic business decision-making. The study involves systematic data preparation, exploratory data analysis (EDA), and the development of interactive visualizations using Power BI.

The analysis focuses on evaluating important aspects such as overall sales performance, regional and category-wise contributions, customer segmentation, and factors influencing profitability. By examining these components, the project provides a clear understanding of how different variables impact business outcomes.

In addition, the project emphasizes the use of data visualization techniques to transform raw data into meaningful insights. An interactive dashboard has been created to enable dynamic exploration of the dataset, allowing users to filter and analyse information efficiently.

The ultimate goal of this project is to leverage data analytics to support better decision-making, improve business strategies, and highlight opportunities for growth and optimization within the retail domain.

2. DATASET DESCRIPTION

The dataset used in this project was sourced from Kaggle and contains detailed retail transaction records.

Key Features:

- **Order Date, Ship Date**
- **Category, Sub-Category, Product Name**
- **Sales, Profit, Quantity, Discount**
- **Region, Segment, Customer Details**

Data Types:

- **Numerical:** Sales, Profit, Quantity, Discount
- **Categorical:** Category, Region, Segment

The dataset is suitable for analysing business performance and identifying key revenue drivers.

3. DATA CLEANING & PREPROCESSING

The dataset was carefully cleaned and pre-processed to ensure accuracy, consistency, and reliability for further analysis. The following steps were performed:

- **Removed duplicate records** to maintain data integrity and avoid redundant entries that could distort analysis results. Duplicate rows were eliminated.
- **Handled missing/null values** by inspecting each column and applying suitable techniques.
- proper datetime format to enable time-based analysis such as monthly trends, seasonal patterns, and time-series visualization. .
- **Created new derived features** to enhance analysis:
    - **Month (MMM YYYY format):** Used for visualizing sales trends over time
    - **Month Sort:** Created to ensure correct chronological order in time-series charts
- **Verified and corrected data types** for all columns (e.g., numerical columns like Sales, Profit, Quantity, Discount were ensured to be in appropriate numeric format).
- **Checked and cleaned categorical data** to remove inconsistencies such as spelling variations or unwanted spaces in fields like Category, Region, and Segment.
- **Validated numerical values** to ensure there were no incorrect or unrealistic entries (e.g., negative quantities or invalid sales values).
- **Prepared the dataset for visualization tools (Power BI)** by structuring it in a clean and analysis-ready format, enabling efficient creation of dashboards and reports.

4. EXPLORATORY DATA ANALYSIS (EDA)

The exploratory data analysis was performed using Power BI to identify patterns, trends, and relationships within the retail dataset. The analysis is based on key visualizations developed in the dashboard.

4.1 Overall Performance (KPI Analysis)

The dashboard highlights the overall business performance using key indicators:

- Total Sales: 742.00K
- Total Profit: 18.45K
- Total Quantity Sold: 8028

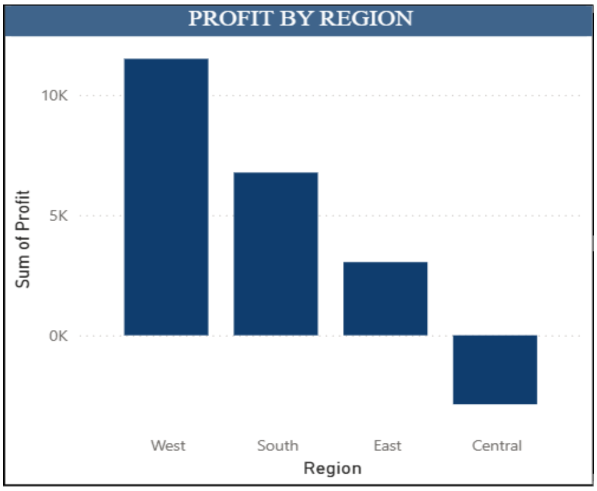
These metrics provide a summary of the business, indicating moderate profitability relative to total sales**.**

4.2 Region-wise Profit Analysis

Profit analysis across regions reveals:

- West region has the highest profit
- South and East regions show moderate profitability
- Central region shows negative profit, indicating losses

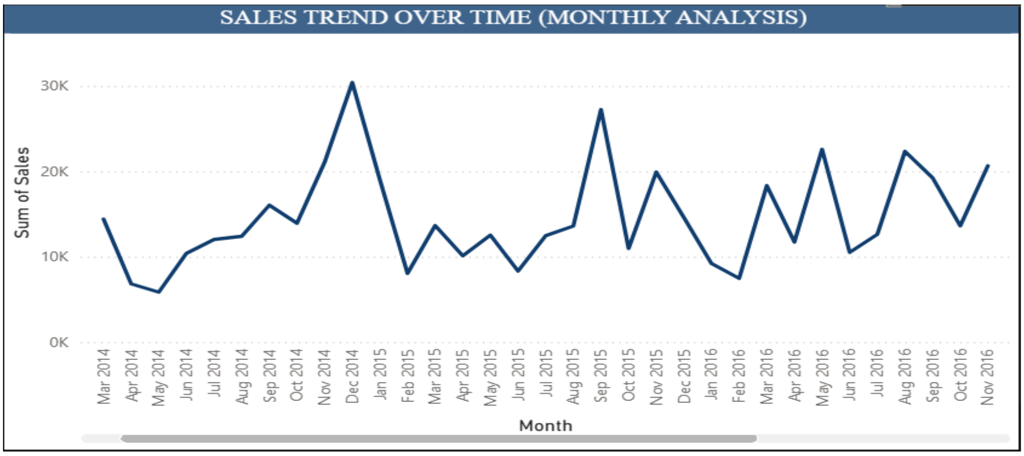
This highlights that high sales do not always guarantee high profit and points to inefficiencies in certain regions.



4.3 Sales Trend Over Time (Monthly Analysis)

The monthly sales trend shows fluctuations over time, indicating variability in sales performance.

- Sales exhibit seasonal patterns, with certain months showing peaks
- A noticeable spike is observed around late 2014 and 2015
- Some months show lower performance, indicating possible off-season periods

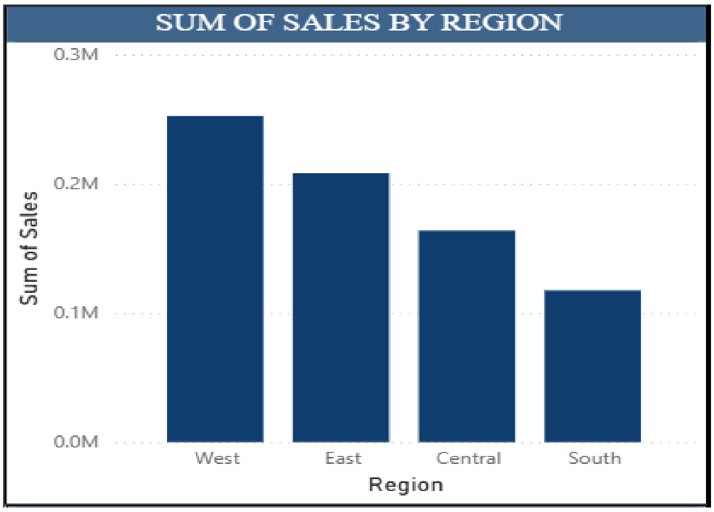


This suggests that sales are influenced by time-based factors such as demand cycles and promotions.

4.4 Region-wise Sales Analysis

The analysis of sales across regions shows:

- West region generates the highest sales
- Followed by East and Central regions
- South region contributes the least

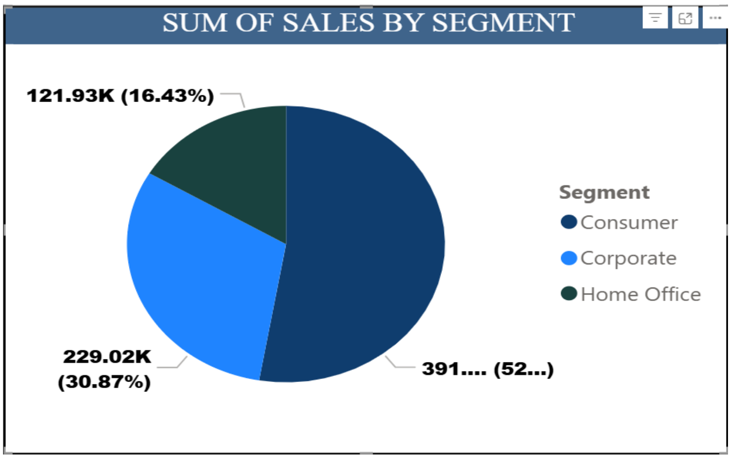


This indicates that the West region is a key revenue contributor and has strong market performance.

4.5 Sales by Customer Segment

The pie chart shows distribution of sales among customer segments:

- Consumer segment contributes the highest share (~52%)
- Corporate segment contributes around 30%
- Home Office segment contributes the least (~16%)

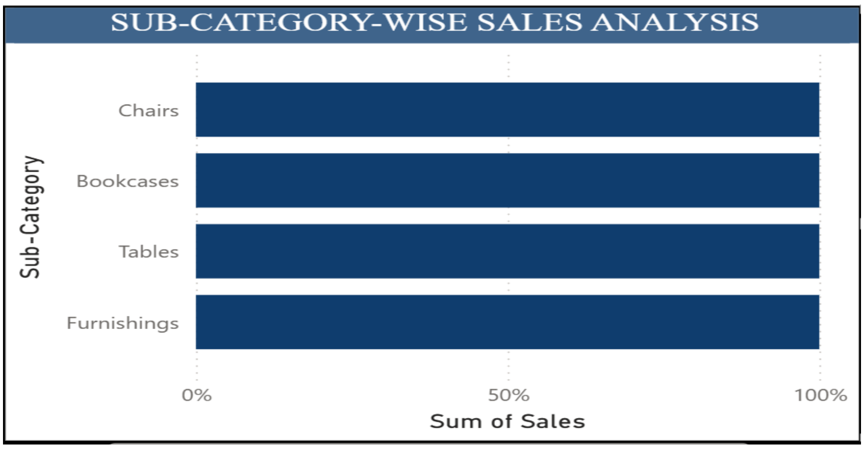


This indicates that individual customers are the primary drivers of revenue.

4.6 Sub-Category-wise Sales Analysis

Sales distribution across sub-categories shows:

- Categories such as Chairs, Tables, Bookcases, and Furnishings contribute significantly
- Sales are relatively evenly distributed among these sub-categories

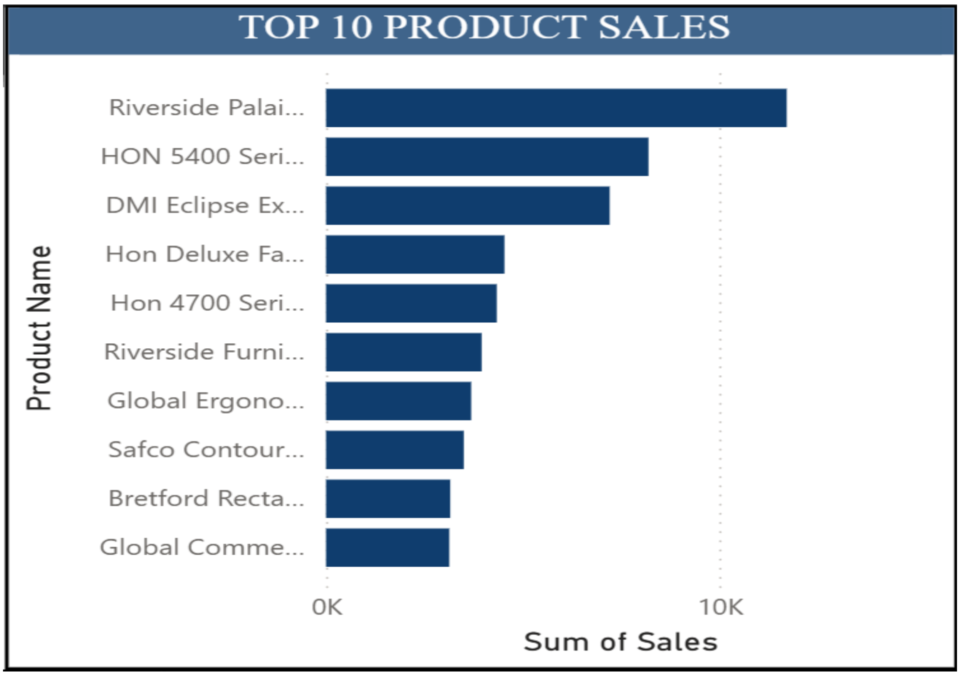


This indicates stable demand across multiple product types within the category.

4.7 Top 10 Products Analysis

The Top 10 products chart highlights:

- A small number of products contribute a large portion of total sales
- Products like HON series items and Riverside products are among top performers followed by DMI Eclipse.



This reflects the importance of focusing on high-performing products for maximizing revenue.

4.8 Dashboard Interactivity

The dashboard includes interactive filters (Region slicer), allowing users to:

- Dynamically explore data
- Compare performance across regions
- Analyze trends for specific selections

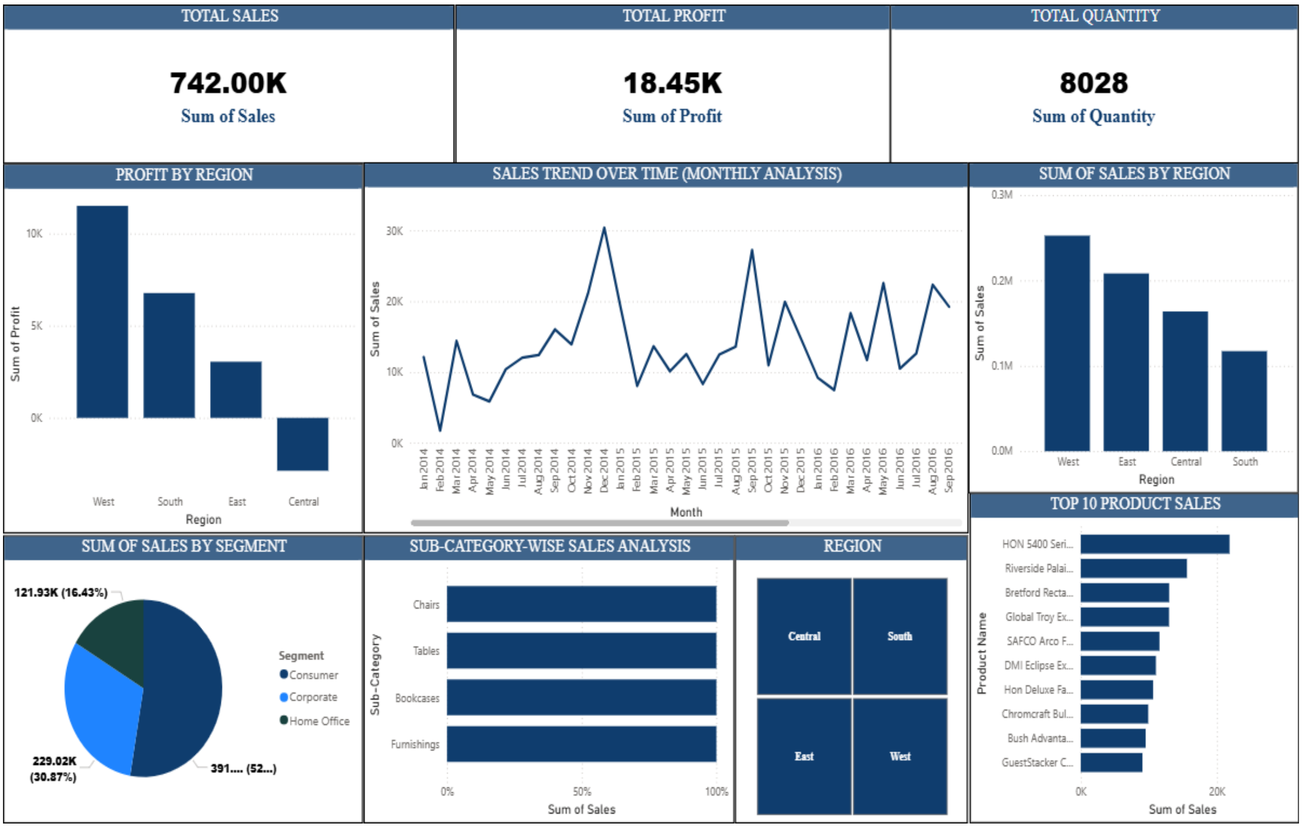
This enhances usability and enables deeper analysis.

4.9 Dashboard Visualization (Power BI)

An interactive Power BI dashboard was developed to visualize insights.

Dashboard includes:

- KPI Cards:
  - Total Sales
  - Total Profit
  - Total Quantity
- Charts:
  - Sales by Category & Sub-category
  - Sales Trend (Monthly)
  - Sales by Region
  - Profit by Region
  - Sales by Segment
  - Top 10 Products
- Filters (Slicers):
  - Category
  - Region



The dashboard allows users to dynamically explore data and gain deeper insights.

5. KEY INSIGHTS

- The overall sales performance indicates that revenue is concentrated in specific product categories, with certain categories consistently contributing a higher share to total sales.
- The **West region emerges as the strongest performing region**, contributing significantly to both sales and profit, while the **Central region shows weaker performance and even negative profitability**, indicating operational challenges.
- Customer behavior analysis reveals that the **Consumer segment is the primary revenue driver**, highlighting the importance of individual customers over corporate and home office segments.
- Sales trends over time indicate noticeable fluctuations, suggesting the presence of **seasonal demand patterns and periodic peaks in sales performance**.
- Profit analysis shows that **higher sales do not always translate into higher profit**, as certain regions and products generate low or negative profit despite strong sales.
- A limited number of products contribute disproportionately to total revenue, indicating a **high dependency on top-performing products**.

- Variations in profitability across regions suggest that **factors such as discount strategies, logistics, and operational costs significantly influence profit margins**.

6. BUSINESS RECOMMENDATIONS

- Focus on **Technology products** to maximize revenue
- Improve performance in **low-performing regions (Central)**
- Optimize discount strategies to avoid profit loss
- Use seasonal trends for better planning
- Focus on top-performing products
- Reduce unnecessary discounts

7. CONCLUSION

This project demonstrates how data analytics can be used to extract meaningful insights from retail data. The findings help in understanding sales trends, customer behavior, and profitability drivers.
The Power BI dashboard provides a clear and interactive way to analyze business performance, supporting better strategic decision-making.

Additionally, the use of visualization tools enhances the ability to interpret complex data in a simple and effective manner. The insights derived from this analysis can assist businesses in optimizing operations, improving customer targeting, and increasing overall profitability.

RETAIL STORE SALES INSIGHTS AND PREDICTION MODEL

PHASE 2

In this phase 2, machine learning models were developed to predict sales using the cleaned dataset prepared in Phase 1. The focus was on understanding how different features influence sales and building models that can provide accurate predictions.

8.1 FEATURE PREPARATION

- Extracted time-based features such as **Year, Month, and Day** from the Order Date column.
- Selected important input features including:
  - Category
  - Sub-Category
  - Region
  - Segment
  - Quantity
  - Discount
- Converted categorical variables into numerical format using **Label Encoding**.
- Prepared the dataset by separating:
  - **Input features (X)**
  - **Target variable (Sales)**

These steps ensured the dataset was ready for machine learning modelling.

8.2 MODEL DEVELOPMENT

Two models were used for sales prediction:

- **Linear Regression**
  - Used as a basic model to understand simple relationships between features and sales.
  - Easy to implement but limited in handling complex patterns.
- **Random Forest Regressor**
  - Used as an advanced model to capture non-linear relationships.
  - Works by combining multiple decision trees for better accuracy.

The dataset was split into **training and testing data (80:20)** to evaluate performance.

8.3 MODEL PERFORMANCE EVALUATION

The models were evaluated using standard metrics:

- **R^2 Score** → Measures how well the model explains sales variation
- **RMSE** → Measures prediction error

Results:

- Linear Regression:
  - R^2 Score: **0.169**
  - RMSE: **504.40**
- Random Forest:
  - R^2 Score: **0.401**
  - RMSE: **428.30**

The Random Forest model performed better with higher accuracy and lower error.

8.4 MODEL COMPARISON

- Linear Regression showed limited performance due to its inability to capture complex patterns.
- Random Forest provided better predictions as it handled non-linear relationships effectively.
- Lower RMSE in Random Forest indicates more reliable predictions.

Therefore, Random Forest was selected as the **best model for this project**.

9. BUSINESS INSIGHTS & RECOMMENDATIONS

Based on both data analysis and model results:

- Sales are influenced by multiple factors such as category, region, and discount.
- High discounts tend to reduce overall profit, so they should be controlled.
- Certain regions perform better and should be prioritized for growth.
- Seasonal trends can be used for better planning and forecasting.
- A small number of products contribute significantly to total sales.
- Machine learning predictions can help in better inventory and demand planning.

10. CONCLUSION

This phase focused on applying machine learning techniques to predict sales and support business decisions. The results show that advanced models like Random Forest perform better than simple models like Linear Regression.

By combining data analysis with predictive modelling, the project provides useful insights that can help improve business performance, optimize strategies, and make better decisions for future growth.